

# Bridging Landau–Levich Theory and Machine Learning for Bayesian-Optimized Thin-Film Thickness Prediction

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## Abstract

Dip coating is a widely used thin-film manufacturing method in biomedical devices, electronics, optics, and other solution-processed technologies, yet bonded-thickness prediction remains difficult because outcomes depend on withdrawal hydrodynamics, solvent chemistry, solids loading, and post-deposition consolidation. Classical relations such as the Landau–Levich formulation are therefore insufficient for multicomponent polymer systems. This study develops a machine-learning framework for predicting bonded thickness in dip-coated poly(dimethylsiloxane) (PDMS) films processed with hexane, toluene, and ethyl acetate. Because experimental data were scarce, synthetic augmentation via bootstrap resampling with Cholesky-correlated noise was used to expand the dataset to 3000 samples across all three solvents, with physicochemical descriptors (polarity, viscosity, boiling point, surface tension) added to the original experimental variables. Five regression families were benchmarked: linear regression,  $k$ -nearest neighbors, backpropagation neural networks, random forest, and gradient boosting. A sequential holdout study revealed severe distribution shift, driving all baseline models to negative  $R^2$ ; stratified 5-fold cross-validation recovered high scores within the synthetic corpus. A Bayesian-optimized random forest achieved a cross-validated search score of  $R^2 = 0.9734$ . Because augmentation and model selection were not isolated from all validation dependencies, this is a development result rather than independent generalization evidence. A target-separated follow-up trained a second Bayesian-optimized random forest on mobile thickness, defined as total minus bonded thickness. It achieved out-of-fold  $R^2 = 0.9996$  on 2887 synthetic rows with nonnegative mobile thickness. On the solvent-specific experimental tables, a concentration-adjusted Landau–Levich estimate described the hexane mobile layer well ( $R^2 = 0.9905$ ) but failed for toluene and ethyl acetate under the current fixed-property assumptions. Landau–Levich was excluded from all bonded-thickness comparisons. The results demonstrate that the dominant challenges were methodological rather than algorithmic: data scarcity, feature incompleteness, and validation design each materially shaped model conclusions. Feature- importance and residual analyses identify concrete directions for improved physicochemical descriptors and synthetic-data generation. An experimental-only follow-up audits 142 source records and retains 126 individual readings for primary validation. Without measured-thickness features, a fixed random forest achieves grouped-CV  $R^2 = 0.568$  for bonded and 0.879 for mobile thickness, but both pooled leave-one-solvent-out scores are negative. High-concentration mobile-layer interval coverage falls to 14.8% at a nominal 80% level. Independent new experiments remain necessary before deployment claims or manuscript finalization.

# 1 Introduction

## 1.1 Importance of dip coating in modern technology

Dip coating is one of the most broadly used wet-processing methods for depositing functional films on solid surfaces. Its importance extends across biomedical devices, drug-delivery and implant coatings, transparent and conductive layers in electronics, dielectric and passivation coatings in semiconductor processing, antireflective and protective optics, corrosion-resistant aerospace surfaces, storage media, flexible sensors, energy devices, filtration membranes, and polymer-based surface modifications. In each of these application areas, film thickness is not a cosmetic detail; it directly influences optical transmission, adhesion, barrier performance, electrical behavior, mechanical durability, biocompatibility, and chemical stability.

The practical appeal of dip coating is clear. It is experimentally simple, adaptable to different geometries, compatible with laboratory and scale-up settings, and naturally suited to solution-processable materials. Yet those same advantages come with complexity. Real dip-coating outcomes depend on the coupled action of fluid viscosity, withdrawal conditions, surface tension, solvent volatility, concentration, substrate condition, and post-withdrawal solidification. In polymer-solvent systems, the final bonded layer can differ substantially from the initial wet film, especially when evaporation, interfacial transport, or bonding phenomena alter the coating after withdrawal.

## 1.2 Classical modeling and its limitations

The classical fluid-mechanics description of plate withdrawal is most commonly associated with the Landau–Levich law and later Landau–Levich–Derjaguin interpretations. For a Newtonian liquid withdrawn at velocity  $U$ , the wet entrained thickness is often expressed as

$$h \approx 0.94 \ell_c \text{Ca}^{2/3}, \quad (1)$$

where  $\ell_c = \sqrt{\gamma/(\rho g)}$  is the capillary length and  $\text{Ca} = \mu U/\gamma$  is the capillary number [1, 2]. This relation is foundational because it formalizes the balance between viscous drag out and capillary restoration. In the present study, it is computed only for the derived mobile-layer comparison and is never treated as a bonded-thickness model.

However, that law does not solve the bonded-thickness problem. It predicts an entrained liquid film under a restricted hydrodynamic regime, whereas bonded thickness is created by interfacial attachment and subsequent film evolution. It also omits chemistry-specific descriptors and evaporation conditions that can change the liquid-to-solid conversion. The present work therefore enforces a strict target boundary: Landau–Levich is not scored against bonded thickness, but it is evaluated numerically against the derived mobile layer. Bonded thickness is compared only with adsorption-motivated analytical models and machine-learning predictions.

For these reasons, classical dip-coating theory should be seen here not as a competing framework but as physical structure for the learning problem. The study is not concerned with replacing coating physics; rather, it develops a data-driven surrogate for bonded thickness and uses Landau–Levich as an uncalibrated, target-appropriate baseline for mobile thickness.

### 1.3 Why machine learning is a strong candidate

Machine learning is especially promising for this class of problem because the governing inputs are heterogeneous, nonlinear, and interaction-heavy. Concentration affects solids loading, solvent properties modulate capillary and evaporation behavior, and thickness-related variables encode process-state information that may summarize prior deposition history. These combinations are difficult to express analytically, especially in a data-limited environment. Machine learning can instead infer the mapping

$$\hat{y} = \hat{f}(x), \quad (2)$$

where  $x$  contains process and physicochemical descriptors and  $y$  is the bonded thickness.

This study therefore asks whether supervised learning can provide a practical predictive route for PDMS dip-coated films when classical theory is informative but incomplete.

The study first documents historical small-data and synthetic benchmarks, sequential holdout failures, descriptor expansion, ensemble comparison and Bayesian model selection. It then separates bonded and mobile targets for analytical comparison. The final experimental-only robustness study audits physical constraints and units, removes all measured-thickness predictors, and evaluates grouped, solvent-transfer and concentration-shift performance. Historical benchmark scores and the new forecasting results answer different questions and are interpreted separately.

## 2 Background and Related Work

### 2.1 Hydrodynamic entrainment and the Landau–Levich–Derjaguin regime

Landau and Levich provided the canonical asymptotic description of liquid entrainment by a moving plate [1]. For a clean Newtonian liquid at small capillary number, matching the dynamic meniscus to the static bath gives

$$h_{\text{LLD}} = 0.94 \ell_c \text{Ca}^{2/3} = 0.94 \frac{(\mu U)^{2/3}}{\gamma^{1/6} (\rho g)^{1/2}}, \quad (3)$$

where  $\mu$  is dynamic viscosity,  $U$  is withdrawal speed,  $\gamma$  is surface tension,  $\rho$  is liquid density, and  $g$  is gravitational acceleration. Wilson’s drag-out analysis clarified the asymptotic structure and the limited range in which the classical approximation is expected to hold [3]. The law describes the liquid film leaving the meniscus; it is not, by itself, a law for the final polymer mass or the chemically bonded fraction.

This distinction becomes decisive for polymer solutions. After withdrawal, solvent loss changes both thickness and composition. If the entrained solution has polymer volume fraction  $\phi_p$ , a first-order dry-film estimate is  $h_{\text{dry}} \approx \phi_p h_{\text{LLD}}$ , assuming complete solvent removal, negligible volume change on mixing, and spatially uniform solids retention. Direct liquid/solid measurements by Zhang, Peng, and Kornev showed that liquid polymer films can follow Landau–Levich–Derjaguin scaling even when the resulting solid-film thickness and morphology do not [8]. Thus a hydrodynamic match at withdrawal does not guarantee a correct post-evaporation prediction.

Reviews of dip coating further document deviations caused by surfactants, Marangoni stresses, non-Newtonian rheology, substrate chemistry, particles, and evaporation [2]. These effects motivate

treating Landau–Levich as a target-specific baseline with explicit assumptions, rather than as a universal coating-thickness equation.

## 2.2 Bonded and mobile layers as distinct physical responses

The most direct precedent for the present target separation is the dip coating of functional lubricant layers studied by Merzlikine et al. [4]. Their measurements attributed the bonded layer to surface adsorption and found it to depend on dwell time, whereas viscous-flow entrainment created a mobile layer governed primarily by pull-out speed. The total deposited film can therefore be represented conceptually as

$$h_T = h_B + h_M, \quad (4)$$

where  $h_B$  is bonded thickness and  $h_M$  is mobile thickness. A model derived for  $h_M$  should not be fitted or scored as though it predicts  $h_B$ .

Adsorption kinetics supply a more appropriate analytical starting point for the bonded component. A reversible site-limited balance can be written as

$$\frac{dh_B}{dt} = k_a C (h_{B,\infty} - h_B) - k_d h_B, \quad (5)$$

which yields a saturating exponential response under fixed conditions. Merzlikine et al. described their bonded-layer observations with reversible adsorption kinetics [4]. The two bonded-layer equations used in this project retain this saturation behavior, but their concentration and measured-film-state terms are fitted, monotonic extensions. They are therefore semi-empirical adsorption-motivated comparators, not first-principles identification of  $k_a$ ,  $k_d$ , or an equilibrium site density.

## 2.3 Evaporation, capillary feeding, and mixed deposition regimes

At sufficiently low withdrawal speed, evaporation near the meniscus can draw solution upward by capillary flow. At higher speed, viscous entrainment and gravitational drainage dominate. Faustini et al. used acceleration-mode dip coating to show how evaporation and viscosity shape thickness profiles across these regimes [5]. A commonly used semi-empirical decomposition expresses the capillary contribution as

$$h_{\text{cap}} \propto \frac{E}{LU}, \quad (6)$$

where  $E$  is an effective evaporation rate and  $L$  is the coating-zone width, while the draining contribution follows the characteristic  $U^{2/3}$  scaling. Their sum produces a mixed regime and the familiar thickness minimum between low- and high-speed limits.

The effective evaporation rate is not simply a tabulated pure-solvent volatility. Bindini et al. demonstrated that solvent partial-pressure gradients and apparatus geometry can alter evaporation and create large thickness variations [6]. Fan and Waltman likewise observed capillary, drainage, and crossover regimes in dip-coated polymer lubricant films, with solvent compatibility and solute concentration affecting which regime appeared [7]. Because the present datasets do not record a local effective  $E$ , the capillary-only and combined equations are retained symbolically rather than assigned an unsupported numerical value.



## 2.4 Implications for analytical benchmarking

The literature establishes three non-interchangeable modeling levels: wet-solution entrainment, mobile polymer remaining after solvent loss, and surface-bonded polymer. This project maps them explicitly. Landau–Levich is converted from wet-solution thickness to a dry mobile-PDMS estimate and compared only with  $h_M$ . Adsorption-motivated models are compared only with  $h_B$ . Evaporation-dependent capillary models remain symbolic until the missing process data are measured. This target discipline avoids obtaining an apparently quantitative result by fitting a hydrodynamic mobile-film law to an unrelated bonded response.

## 2.5 Machine learning in materials informatics

Machine learning has become an increasingly important tool in materials science because it can exploit high-dimensional descriptor spaces and uncover nonlinear process–structure or process–property relationships that are difficult to formulate analytically [9, 10]. In thin-film and coating settings, the appeal is obvious: descriptors may include concentrations, thermal properties, rheological measures, surface energetics, and experimentally measured state variables, all of which may interact nonlinearly.

## 2.6 Relevant regression families

The study examined five model families spanning distinct inductive biases. Linear regression serves as a transparent affine baseline.  $k$ -nearest neighbors (KNN) is a local nonparametric model based on geometric similarity [11]. Backpropagation neural networks (BPNN) are multilayer feed-forward models optimized via gradient-based error propagation [12]. Random forest regression (RFR) is a bagged ensemble of decorrelated trees [13]. Finally, gradient boosting regression (GBR) is a stagewise additive tree ensemble that learns by residual correction [14].

## 2.7 Bayesian optimization

Bayesian optimization was used for historical model selection. It treats model validation performance as a black-box objective and adaptively samples promising regions of the hyperparameter space rather than evaluating a fixed exhaustive grid [16]. A search score alone does not establish a predictive ceiling or unbiased generalization performance; that requires evaluation independent of model selection and data generation.

# 3 Materials, Data, and Descriptor Construction

## 3.1 Experimental system

The physical system studied here was PDMS deposited by dip coating using three solvent families: hexane, toluene, and ethyl acetate.

PDMS was treated as the constant solute, while solvent identity changed the physicochemical environment. The original direct experimental modeling effort was conducted on the purely

experimental hexane dataset, using measured concentration, uncoated layer thickness, total thickness, and bonded thickness. The original hexane dataset contains 58 observations and 4 columns. Because 58 samples were insufficient for stable nonlinear regression, direct modeling on the raw experimental table produced trivial or weakly informative outcomes.

### 3.2 Motivation for synthetic-data augmentation

The sparse original dataset created two problems. First, complex learners such as neural networks or flexible tree ensembles could not be reliably compared on only a few dozen observations. Second, the available experimental coverage did not span enough combinations of concentration and thickness states to support a validation workflow. Synthetic-data augmentation was therefore introduced as a methodological necessity.

### 3.3 Stage 1: conditional tabular generative adversarial network synthesis

The first synthetic-data stage used a *conditional tabular generative adversarial network* (CTGAN) through the SDV framework to generate tabular samples from the original experimental table [15]. Because the source table contained only 58 observations, the generator lacked sufficient training signal to produce reliably calibrated samples; the resulting synthetic set was used for initial exploration but was not carried forward as a primary benchmark. This limitation motivated the development of the more conservative augmentation strategy described in Section 3.4.

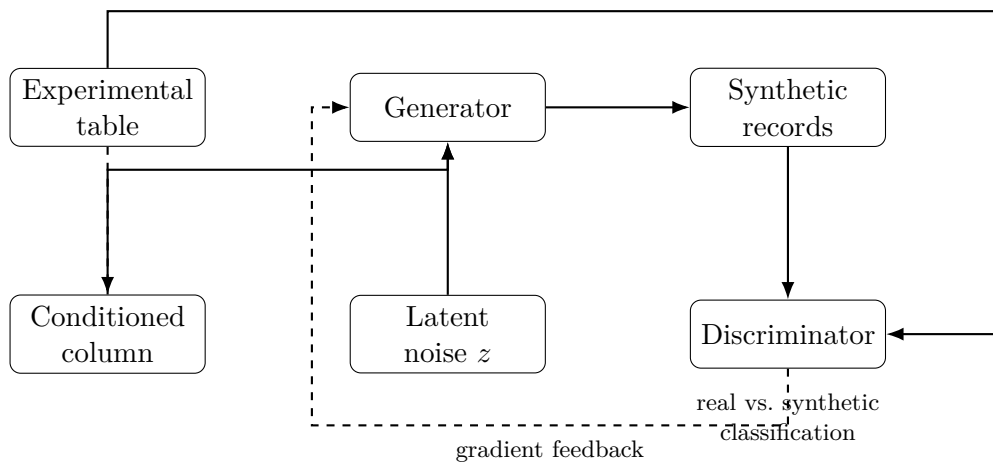


Figure 1: Conceptual architecture of the conditional tabular generative adversarial network used in the first synthesis stage. A conditioned column from the experimental table and latent noise are passed to the generator, while the discriminator learns to separate real and synthetic tabular records and provides adversarial feedback.

### 3.4 Stage 2: bootstrap plus correlated noise

To improve reliability, a second synthetic-data generator was developed. This method bootstrapped rows from the original experimental dataset, estimated the empirical correlation matrix, used Cholesky decomposition to construct correlated perturbations, and added scaled correlated noise

to bootstrapped samples while clipping values near the original data range. This produced a 1000-sample dataset that retained the original four-column structure: concentration (g/mL), uncoated layer (nm), total thickness (nm), and bonded thickness (nm). This dataset was used for the first substantive machine-learning comparison under random train/test splitting.

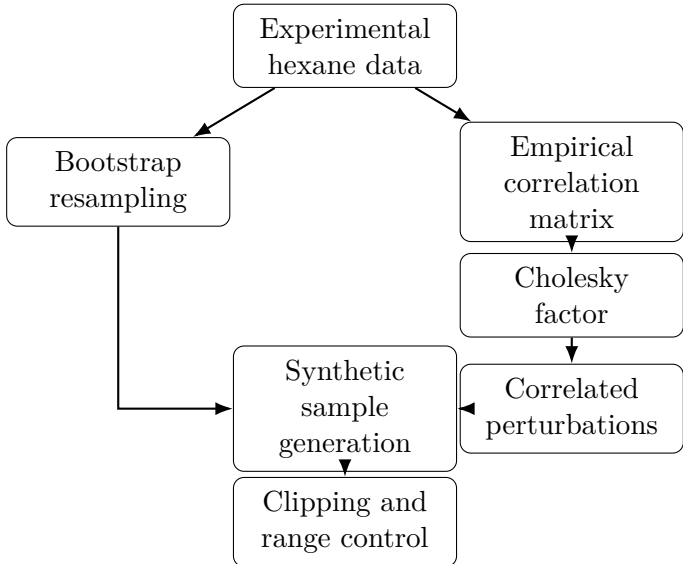


Figure 2: Bootstrap-plus-Cholesky synthesis used in the second augmentation stage. Experimental rows are resampled, the empirical correlation structure is factorized, correlated perturbations are generated, and the final synthetic records are range-controlled to preserve physical plausibility.

### 3.5 Stage 3: feature engineering and solvent-property expansion

To move beyond a purely thickness-and-concentration formulation, the descriptor space was expanded using solvent-property variables collected from PubChem[17]. Four additional descriptors were introduced: polarity (XLogP3), viscosity (cP), boiling point (K), and surface tension (mN/m). The same process was then applied to the hexane, toluene, and ethyl acetate solvent systems. Each solvent-specific table was synthesized to 1000 samples, and the three solvent families were aggregated into a final 3000-sample dataset. The resulting dataset contained both solvent physicochemical descriptors and process-dependent film characteristics, including concentration (g/mL), uncoated layer thickness (nm), total thickness (nm), and bonded thickness (nm).

### 3.6 Dataset summary

Table 1 summarizes the dataset evolution.

Table 1: Datasets used in the investigation.

| Dataset stage                                   | Samples | Variables                 | Role                                                                                                                                                            |
|-------------------------------------------------|---------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Experimental hexane dataset                     | 58      | 3 inputs<br>+ 1<br>target | Original purely experimental dataset used for baseline learning and as the source distribution for synthetic-data generation.                                   |
| Correlation-preserving synthetic hexane dataset | 1000    | 3 inputs<br>+ 1<br>target | Synthetic benchmark set used for preliminary comparison and initial model optimization.                                                                         |
| Aggregated solvent-expanded dataset             | 3000    | 7 inputs<br>+ 1<br>target | Combined hexane, toluene, and ethyl acetate synthetic dataset used for holdout validation, full-dataset cross-validation, ensembles, and Bayesian optimization. |

### 3.7 Statistical characteristics of the aggregated dataset

The aggregated 3000-sample dataset had mean bonded thickness of approximately 0.525 nm and standard deviation approximately 0.183 nm. Concentration and total thickness were the strongest positively associated variables in the final dataset, while polarity and viscosity showed weak negative correlations with the target. Because the synthetic generator allowed limited extrapolation beyond the empirical range, small negative values appear in concentration and total thickness. These values are not physically ideal and are treated as a known limitation of the synthetic-generation stage.

## 4 Model Families and Their Operating Principles

### 4.1 Linear regression

Linear regression assumes that bonded thickness can be expressed as an affine function of the inputs:

$$\hat{y} = \beta_0 + \sum_{j=1}^p \beta_j x_j. \quad (7)$$

Its value in the present investigation is not its flexibility but its interpretability. If a linear model were competitive, then the nonlinear machinery of forests or neural networks would be unnecessary.



Figure 3: Schematic of linear regression as an affine mapping from engineered inputs to predicted bonded thickness.

## 4.2 K-nearest neighbors

KNN regression predicts from the local neighborhood of a query point:

$$\hat{y}(x) = \frac{\sum_{i \in \mathcal{N}_k(x)} w_i y_i}{\sum_{i \in \mathcal{N}_k(x)} w_i}, \quad (8)$$

where the neighborhood  $\mathcal{N}_k(x)$  is determined by a distance metric [11]. KNN can represent non-linear local structure, but it is highly sensitive to how well training and test data occupy the same geometric region.

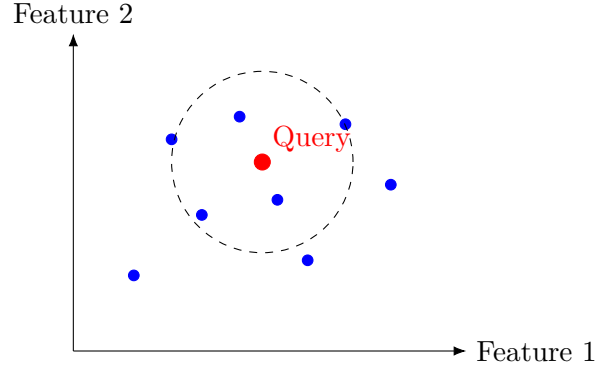


Figure 4: KNN prediction concept: the query point is predicted from nearby training observations inside a local neighborhood.

## 4.3 Backpropagation neural network

The BPNN is a multilayer feed-forward model:

$$\hat{y} = W_L \sigma(W_{L-1} \sigma(\cdots \sigma(W_1 x + b_1)) + b_{L-1}) + b_L, \quad (9)$$

with parameters optimized by backpropagation [12]. BPNNs are expressive but data-hungry and can become unstable when the available training distribution is too small or too shifted.

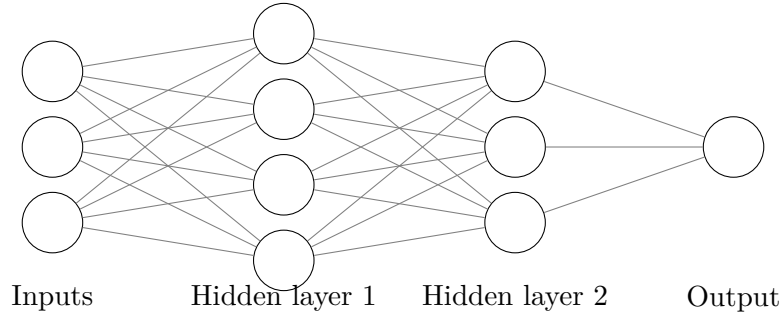


Figure 5: Illustrative multilayer feed-forward neural-network topology for BPNN regression.

## 4.4 Random forest regression

Random forest regression averages many decorrelated trees:

$$\hat{y}(x) = \frac{1}{T} \sum_{t=1}^T f_t(x). \quad (10)$$

Each tree is trained on a bootstrap sample and split using random feature subsets, reducing variance while preserving nonlinear flexibility [13].

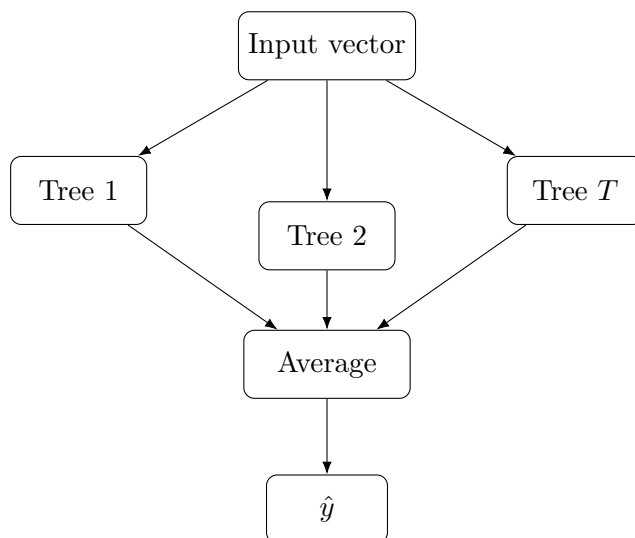


Figure 6: Random forest as an ensemble average of many decorrelated decision trees.

## 4.5 Gradient boosting regression

Gradient boosting builds an additive model stage by stage:

$$F_M(x) = \sum_{m=1}^M \nu \gamma_m h_m(x), \quad (11)$$

where each new learner is fitted to reduce prior residuals [14]. Boosting can be highly accurate but tends to be more hyperparameter-sensitive than bagged forests.

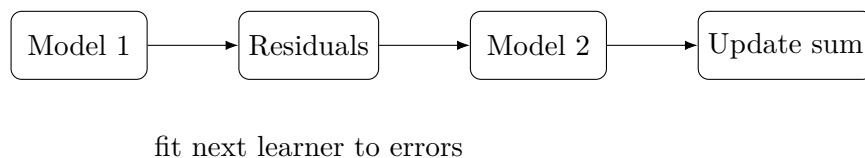


Figure 7: Gradient boosting builds a predictor stagewise by fitting new learners to prior residuals.

## 5 Methodology

### 5.1 Prediction target

The primary prediction target was bonded thickness (nm). This choice is significant because bonded thickness is the experimentally relevant outcome for many PDMS film applications and is not fully determined by the wet-film thickness predicted by classical withdrawal theory. The analytical follow-up introduced a second, separately evaluated target,

$$h_M = h_T - h_B, \quad (12)$$

where  $h_M$ ,  $h_T$ , and  $h_B$  denote mobile, total, and bonded thickness, respectively. Synthetic rows for which this difference was negative were excluded from mobile-layer training. For the solvent-specific experimental comparison, the difference was clipped at zero only when measurement rounding produced a negative value; this affected one toluene row with a raw difference of  $-0.01$  nm. Landau–Levich does not appear in any bonded-thickness metric table.

### 5.2 Evaluation metrics

Performance was reported using mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination ( $R^2$ ):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (13)$$

$$\text{RMSE} = \sqrt{\text{MSE}}, \quad (14)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (15)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (16)$$

### 5.3 Phase I: small-data and synthetic-data benchmarking

The first quantitative phase used the 1000-sample correlation-preserving hexane synthetic dataset. All major model families were trained using random 80/20 train/test splitting, with standard scaling applied to neural-network and distance-based models.

### 5.4 Phase II: sequential holdout validation

The next phase used the 3000-sample aggregated dataset and intentionally adopted a deterministic sequential split:

- rows 0–1999: training set,
- rows 2000–2999: validation set.

This design was used to test whether models trained in one regime generalized to later data without any shuffling.

### 5.5 Phase III: full-dataset stratified cross-validation

Because the sequential split exposed severe distribution shift, the study then moved to a more balanced validation protocol. Continuous bonded-thickness values were binned into target quantiles, and stratified 5-fold cross-validation was performed using those bins. This regression-oriented stratification scheme ensured that each fold saw a similar target-range composition.

### 5.6 Phase IV: ensemble comparison

Two ensemble strategies were evaluated on the full 3000-sample dataset: a fixed 50/50 average of random forest and gradient boosting predictions, and an adaptive blend with inner cross-validation selecting the random-forest weight over the grid  $\{0.0, 0.1, \dots, 1.0\}$ . In addition, a stacked ensemble was trained using ridge regression as a meta-learner over out-of-fold predictions from random forest and gradient boosting.

### 5.7 Phase V: Bayesian optimization

The strongest single model family in the historical benchmark was random forest regression. It therefore entered a model-selection stage, where `BayesSearchCV` optimized:

- `n_estimators` from 100 to 450,
- `max_depth` from 8 to 30,
- `min_samples_split` from 2 to 12,
- `min_samples_leaf` from 1 to 6,
- `max_features` in  $\{\text{sqrt}, \text{log2}, \text{None}\}$ ,
- `bootstrap` in  $\{\text{True}, \text{False}\}$ .

The Bayesian search ran for 32 iterations over the same stratified 5-fold splits used in the full-dataset validation stage.

### 5.8 Phase VI: mobile-layer optimization and analytical comparison

The mobile-layer random forest used the same seven predictors as the final bonded model: polarity, viscosity, boiling point, surface tension, concentration, uncoated-layer thickness, and total thickness. Bonded thickness was used only to construct the response and was not supplied as an input feature. Deriving  $h_M = h_T - h_B$  exposed 113 synthetic rows with negative mobile thickness. Those physically invalid rows were excluded from this target-specific fit, leaving 2887 rows; the tracked source dataset was not altered. Bayesian optimization then repeated the 32-iteration search and target-quantile-stratified 5-fold validation with random seed 42.



For the analytical baseline, the source workbooks identify concentration in g/L even though derived project tables retain the legacy header **Concentration** (g/mL). Concentration was therefore converted as  $c_{\text{g/mL}} = 10^{-3}c_{\text{g/L}}$ , and the PDMS volume fraction was approximated by

$$\phi_{\text{PDMS}} = \frac{c_{\text{g/mL}}}{c_{\text{g/mL}} + \rho_{\text{PDMS}}}, \quad \rho_{\text{PDMS}} = 0.965 \text{ g/mL}. \quad (17)$$

The predicted dry mobile thickness was then

$$\hat{h}_{M,\text{LLD}} = \phi_{\text{PDMS}} h_{\text{LLD}}. \quad (18)$$

This calculation used the documented solvent viscosity, surface tension, and density with a fixed withdrawal speed of 1.0 mm/s. No retention coefficient or bonded-thickness parameter was fitted. The optimized random forest and this uncalibrated analytical baseline were then evaluated separately on the hexane ( $n = 58$ ), toluene ( $n = 51$ ), and ethyl acetate ( $n = 33$ ) experimental tables.

## 6 Results

### 6.1 Preliminary random-split benchmarking on the 1000-sample dataset

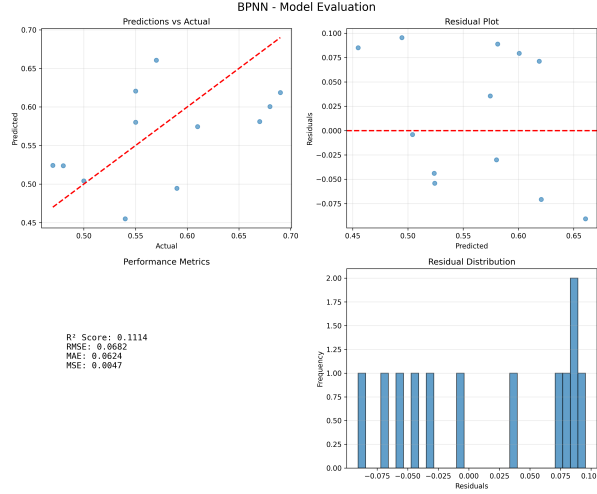
Table 2 summarizes optimized model performance on the 1000-sample correlation-preserving synthetic hexane dataset. Gradient boosting regression was not included in this phase; it was introduced in Phase III alongside the full solvent-expanded feature set, where the seven-variable descriptor space better justified the additional model complexity.

Table 2: Optimized model performance on the 1000-sample synthetic hexane dataset under an 80/20 random split.

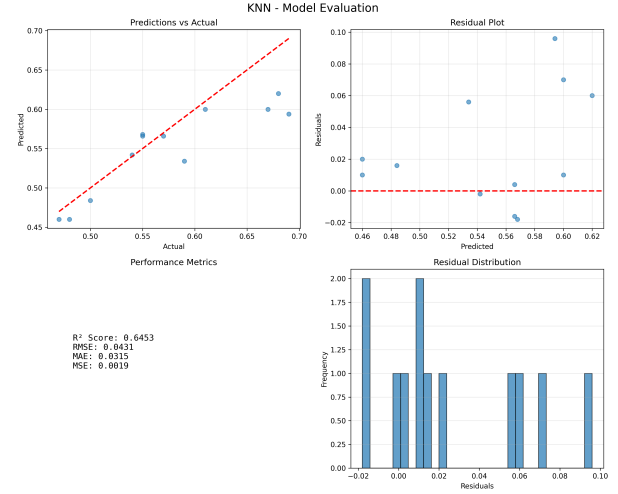
| Model             | $R^2$  | RMSE   | MAE    | MSE     |
|-------------------|--------|--------|--------|---------|
| Random Forest     | 0.8082 | 0.0409 | 0.0273 | 0.00167 |
| BPNN              | 0.7984 | 0.0419 | 0.0311 | 0.00176 |
| KNN               | 0.7972 | 0.0420 | 0.0279 | 0.00176 |
| Linear Regression | 0.7398 | 0.0476 | 0.0399 | 0.00227 |

At this stage, random forest already emerged as the strongest single model, but the performance gap relative to BPNN and KNN was still narrow enough to justify a broader family comparison.

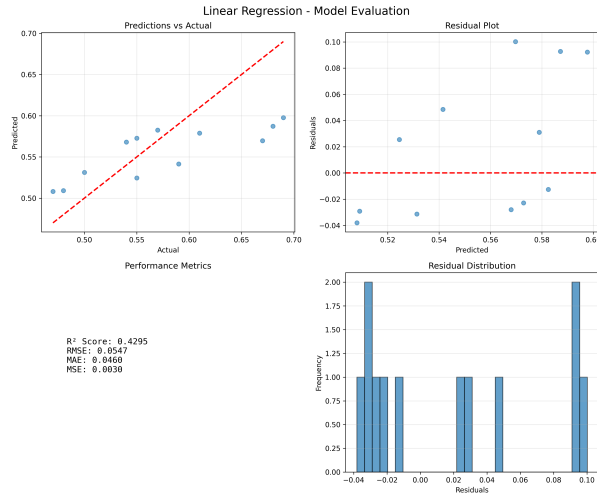
Direct training on the purely experimental hexane dataset preceded the synthetic-data benchmarks and produced the baseline diagnostic plots shown in Figure 8. Those outputs were useful primarily as evidence that the raw experimental table was too small for stable comparative learning.



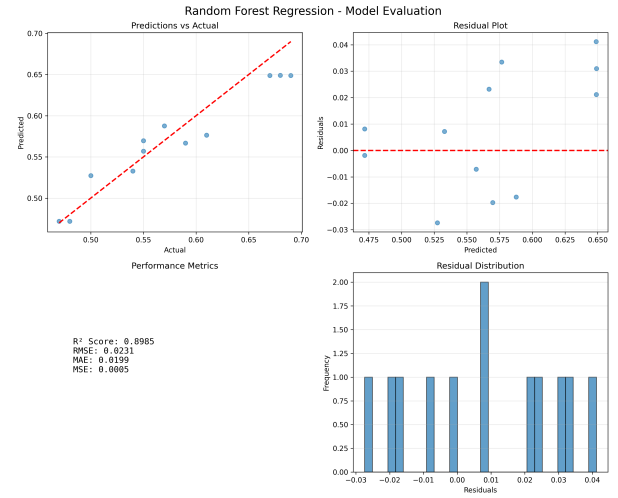
(a) BPNN



(b) KNN



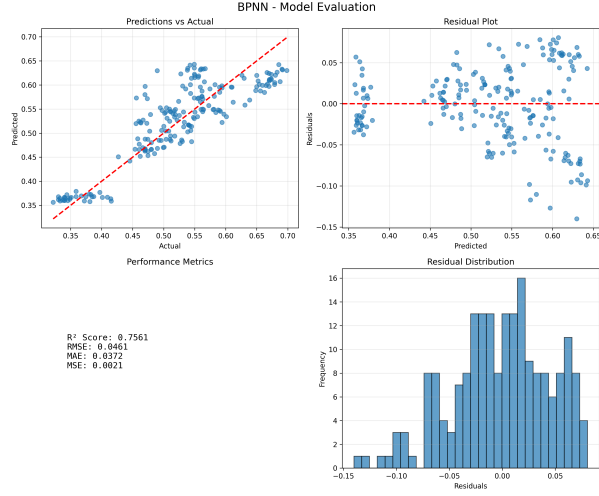
(c) Linear regression



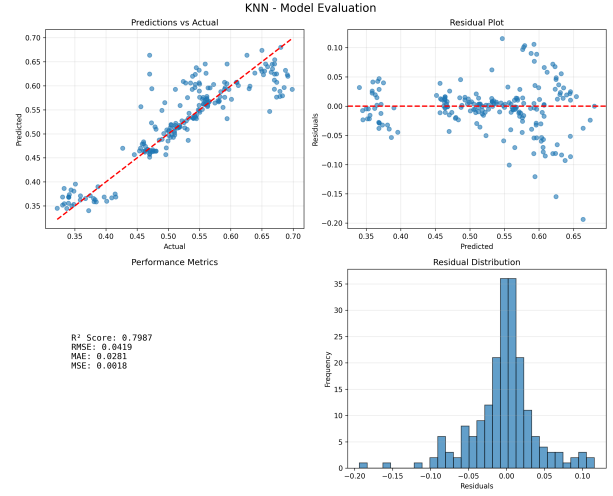
(d) Random forest

Figure 8: Baseline model outputs from the direct experimental hexane dataset. These figures illustrate the limited stability achievable before synthetic-data augmentation.

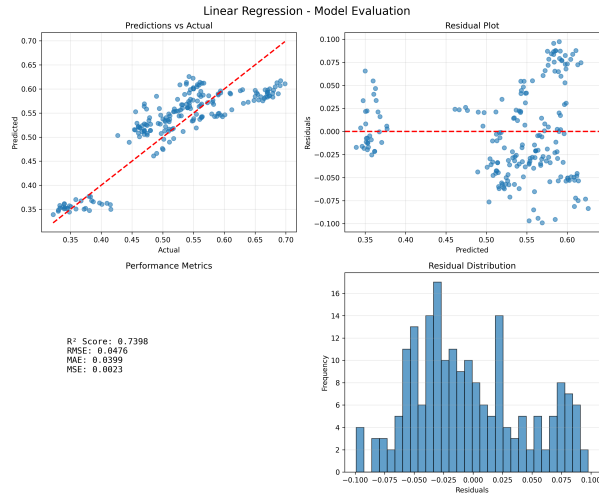
The first correlation-preserving synthetic hexane benchmark produced the evaluation outputs shown in Figure 9, followed by the optimized-model outputs in Figure 10.



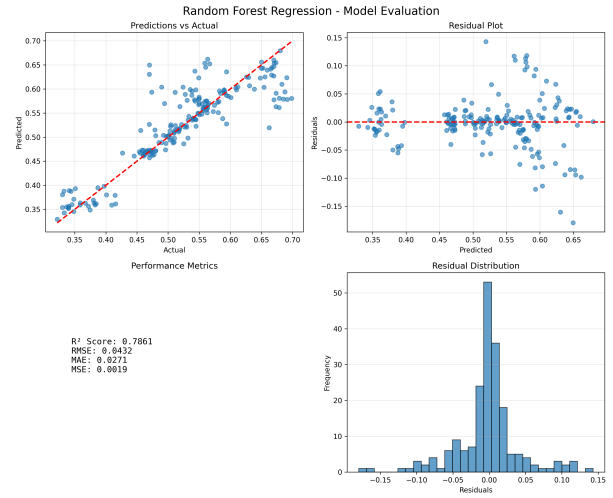
(a) BPNN



(b) KNN

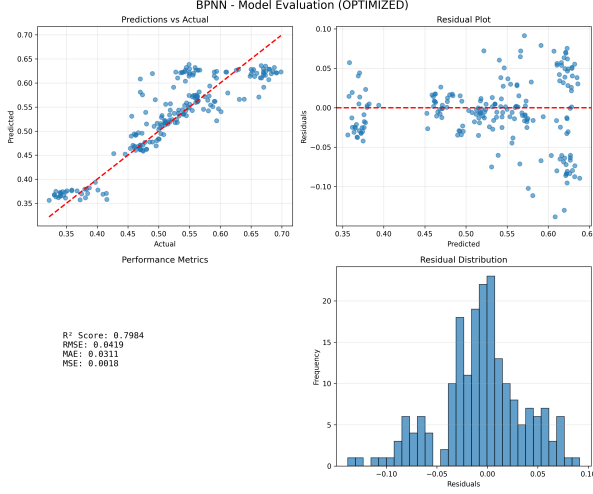


(c) Linear regression

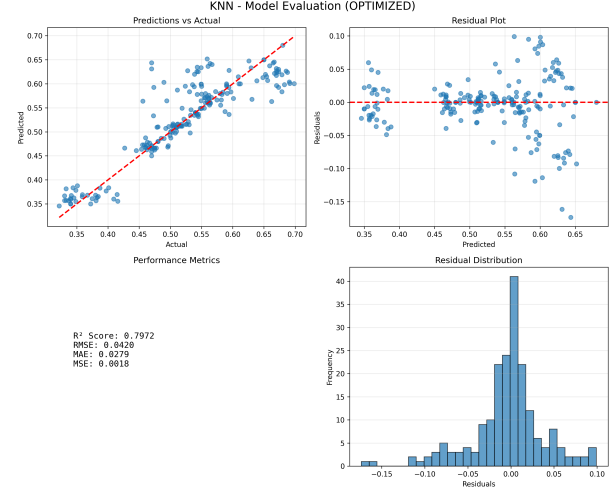


(d) Random forest

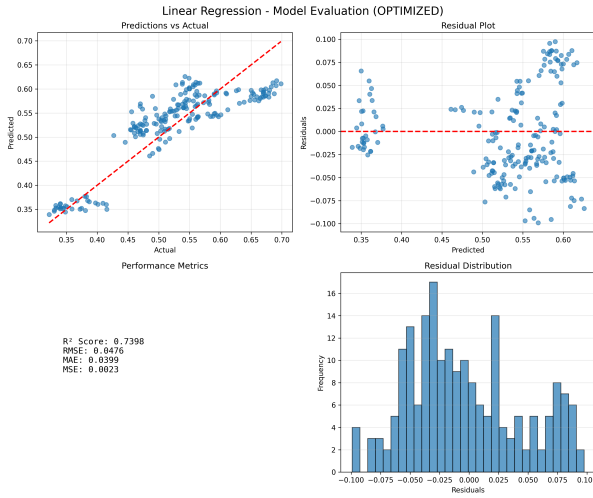
Figure 9: Model outputs on the 1000-sample correlation-preserving synthetic hexane dataset before the later optimization pass.



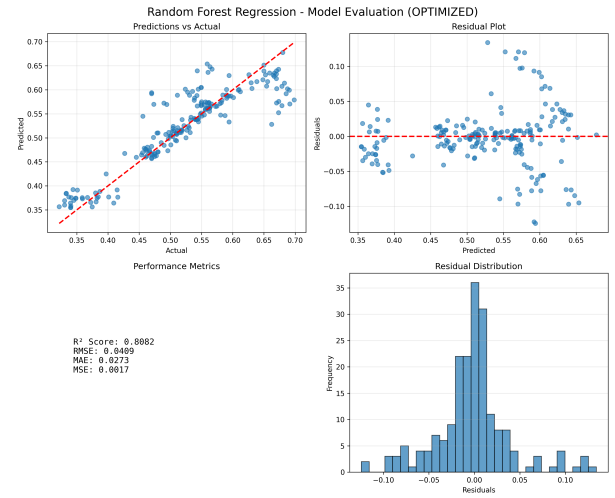
(a) BPNN



(b) KNN



(c) Linear regression



(d) Random forest

Figure 10: Optimized-model outputs on the 1000-sample synthetic hexane dataset. These figures correspond to the results summarized in Table 2.

## 6.2 Sequential holdout validation revealed severe distribution shift

On the 3000-sample aggregated dataset, the sequential holdout baseline was highly unstable. The original 3-feature models performed worse than predicting the mean:

- BPNN:  $R^2 = -331.96$
- Linear regression:  $R^2 = -220.36$
- KNN:  $R^2 = -2.23$
- Random forest:  $R^2 = -1.221$

This collapse was traced to strong regime shift between the first 2000 and last 1000 samples. Concentration, total thickness, surface tension, and viscosity all shifted materially, meaning the validation block occupied a different operating regime than the training block.

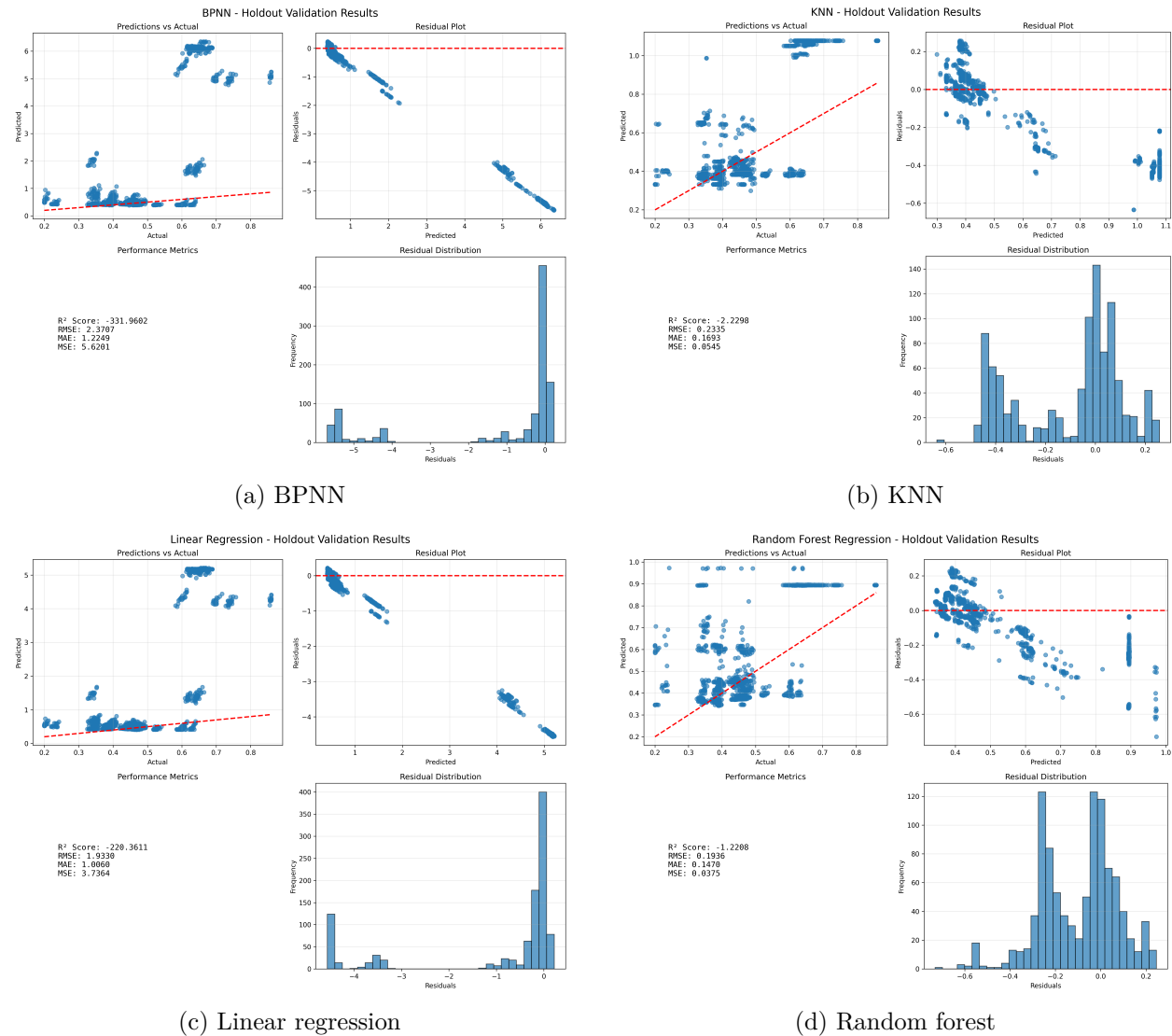


Figure 11: Original sequential holdout-validation outputs on the aggregated 3000-sample solvent-expanded dataset. These figures visually confirm the severe degradation caused by regime shift.

### 6.3 Optimized holdout models after feature expansion

Once all seven descriptors were used and the model focus shifted toward tree ensembles, performance improved sharply. The optimized holdout random forest reached

$$R^2 = 0.3284, \quad \text{RMSE} = 0.1065, \quad \text{MAE} = 0.0745. \quad (19)$$

This was a major recovery, but it still represented a regime-shift validation outcome rather than

a balanced full-distribution estimate.

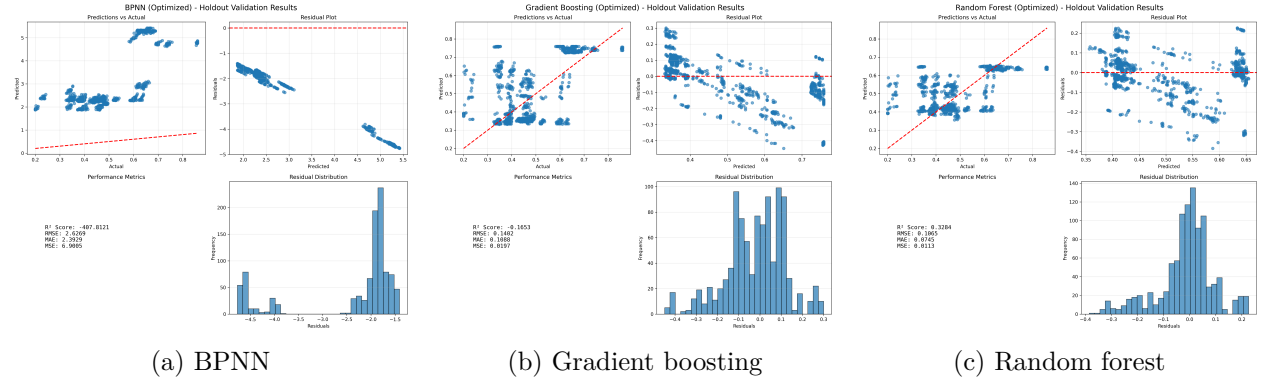


Figure 12: Optimized holdout-validation outputs after feature expansion to seven variables. The random-forest panel corresponds to the strongest regime-shift result.

For completeness, the comparison study that directly juxtaposed original and optimized variants is shown in Figures 13 and 14.

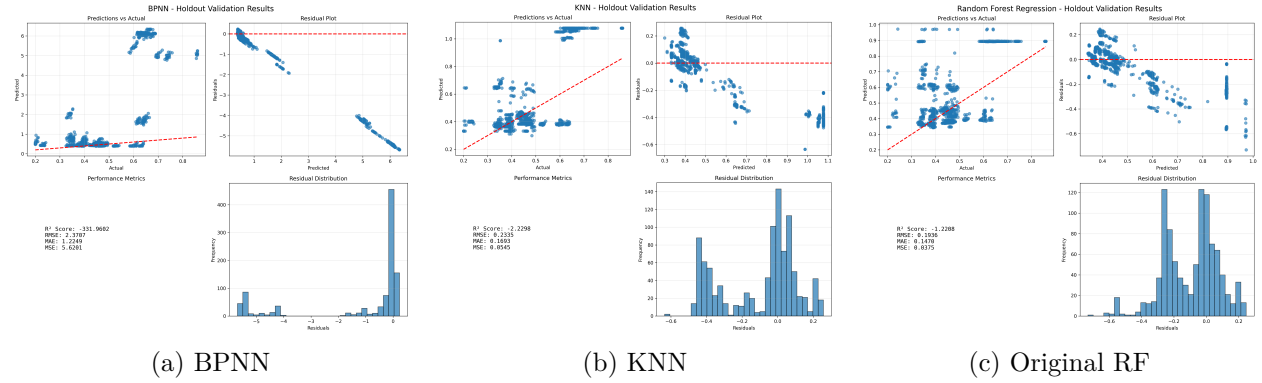


Figure 13: First comparison panel for the holdout study, showing original-model behavior under the sequential validation split.

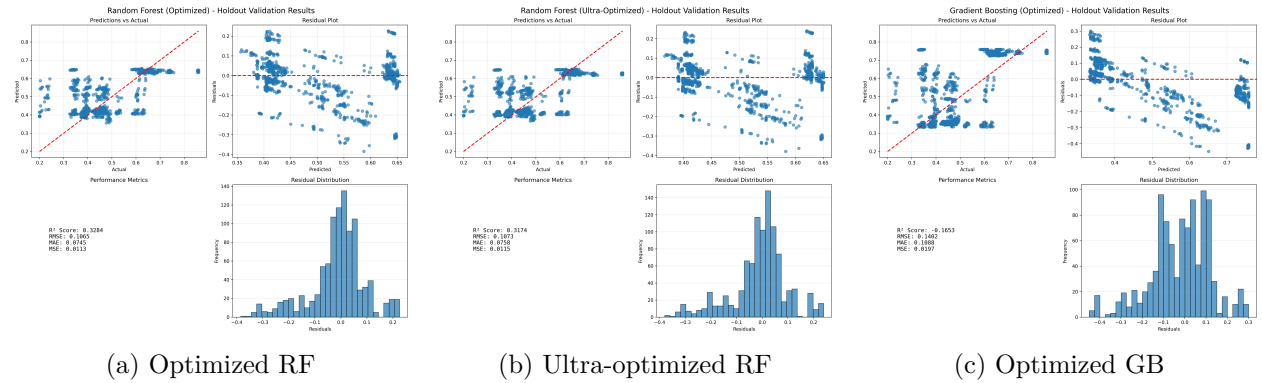


Figure 14: Second comparison panel for the holdout study, highlighting the recovery obtained through feature expansion and tree-based optimization.

## 6.4 Stratified 5-fold cross-validation on the full dataset

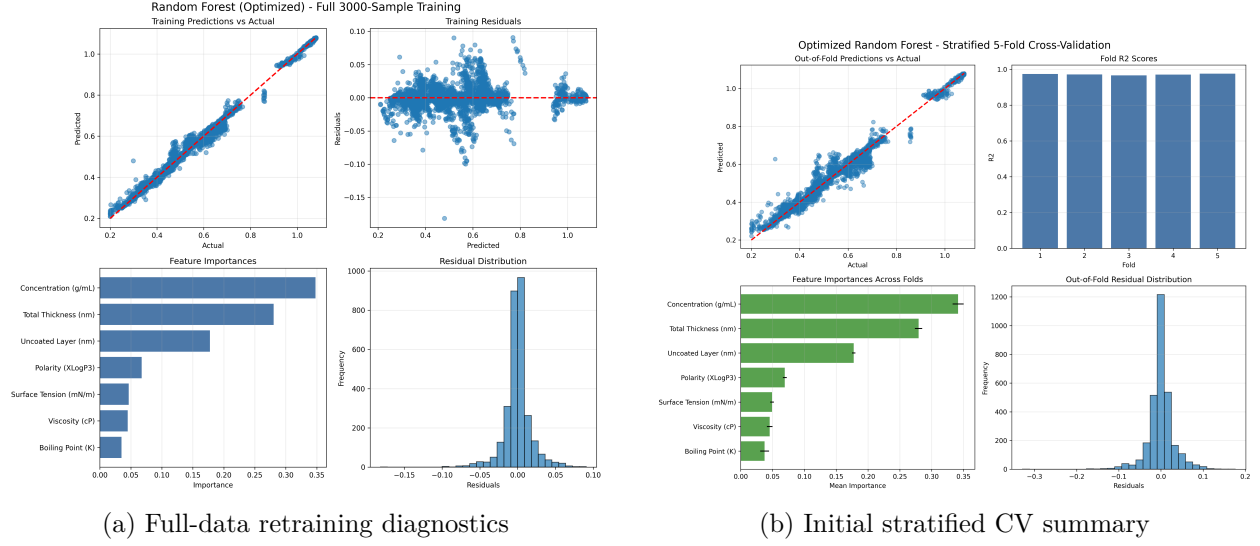
When the same 3000-sample dataset was evaluated with target-stratified 5-fold cross-validation, the optimized random forest achieved:

$$R_{\text{OOF}}^2 = 0.971620, \quad \text{RMSE}_{\text{OOF}} = 0.030851, \quad \text{MAE}_{\text{OOF}} = 0.019663. \quad (20)$$

The mean fold-level  $R^2$  was

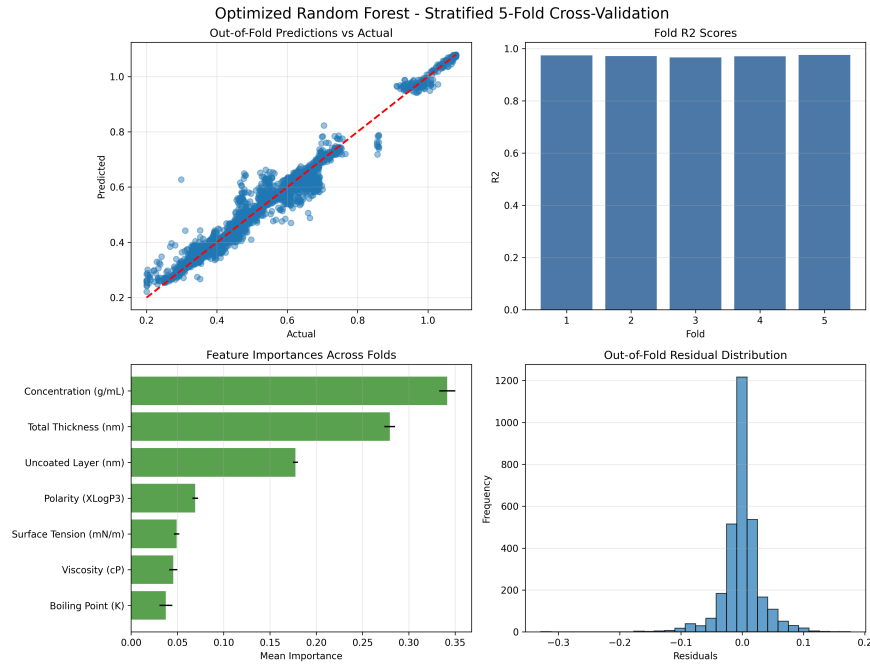
$$R_{\text{mean}}^2 = 0.971593 \pm 0.003533. \quad (21)$$

This result showed that the underlying full-dataset learning problem was highly tractable once sequential partition artifacts were removed.



(a) Full-data retraining diagnostics

(b) Initial stratified CV summary



(c) Corrected stratified CV summary

Figure 15: Full-dataset optimized random-forest diagnostics and stratified cross-validation summaries. The corrected rerun reflects the final metadata-consistent version used for interpretation.

## 6.5 Ensemble comparison

Table 3 compares full-dataset out-of-fold performance for the base models, ensembles, and the Bayesian-optimized random forest.



Table 3: Out-of-fold performance under stratified 5-fold cross-validation for base models, ensembles, and the Bayesian-optimized model.

| Model                          | $R^2_{\text{OOF}}$ | RMSE <sub>OOF</sub> | MAE <sub>OOF</sub> | MSE <sub>OOF</sub> |
|--------------------------------|--------------------|---------------------|--------------------|--------------------|
| <b>Bayesian RF (Optimized)</b> | <b>0.973424</b>    | <b>0.029855</b>     | <b>0.018591</b>    | <b>0.0008914</b>   |
| Stacked Ridge (RF + GB)        | 0.972202           | 0.030533            | 0.019859           | 0.0009323          |
| Adaptive Blend (RF + GB)       | 0.971985           | 0.030652            | 0.019823           | 0.0009395          |
| Fixed Blend 50/50              | 0.971920           | 0.030687            | 0.020147           | 0.0009417          |
| Random Forest                  | 0.971620           | 0.030851            | 0.019663           | 0.0009518          |
| Gradient Boosting              | 0.969843           | 0.031802            | 0.021518           | 0.0010114          |

The Bayesian-optimized random forest achieved the strongest result across all models and ensembles. The stacked ensemble provided the best pure ensemble result, but the improvement over the untuned optimized random forest was modest. Inner cross-validation for the adaptive blend selected random-forest weights between 0.6 and 0.8 across folds, showing that random forest remained the dominant component.

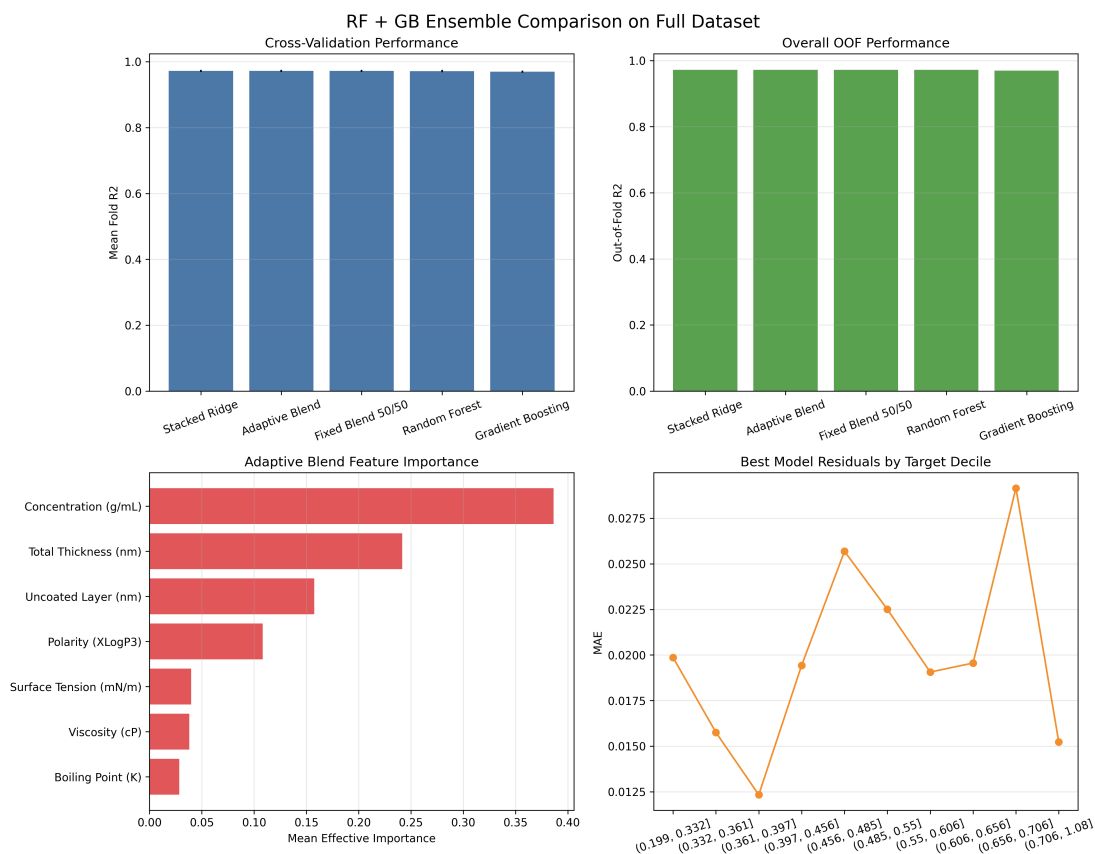


Figure 16: Ensemble-comparison summary figure for random forest, gradient boosting, and RF+GB combinations under stratified 5-fold validation.

## 6.6 Historical Bayesian model-selection performance

Bayesian optimization of random forest regression yielded the following best parameters:

- `bootstrap = True`
- `max_depth = 30`
- `max_features = log2`
- `min_samples_leaf = 1`
- `min_samples_split = 2`
- `n_estimators = 323`

The tuned random forest achieved:

$$R_{\text{OOF}}^2 = 0.973424, \quad \text{RMSE}_{\text{OOF}} = 0.029855, \quad \text{MAE}_{\text{OOF}} = 0.018591. \quad (22)$$

Its best cross-validated search score was  $R^2 = 0.973392$ . When refit on all 3000 samples, the optimized model achieved in-sample  $R^2 = 0.996519$ . The search folds also informed hyperparameter selection, and the synthetic corpus was derived from the source experiments before splitting. These scores therefore describe model development within that corpus and are not an unbiased external generalization estimate. The selected model surpassed the stacked ensemble under the historical protocol; the experimental-only follow-up in Section 7 evaluates a different, more demanding forecasting task.

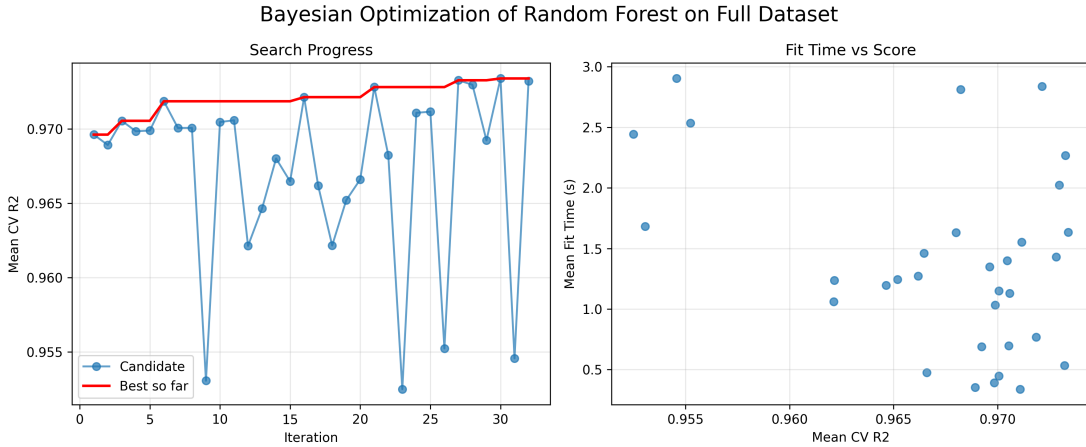


Figure 17: Bayesian optimization summary for the final full-dataset random-forest search. The trace shows progressive improvement in cross-validated  $R^2$  over the search iterations.

## 6.7 Mobile-layer random forest and Landau–Levich comparison

The mobile-target search selected a 450-tree bootstrapped random forest with unrestricted feature sampling, maximum depth 27, minimum leaf size 1, and minimum split size 2. Across the 2887

valid synthetic rows, the baseline random forest achieved out-of-fold  $R^2 = 0.999350$  and RMSE = 0.046171 nm. Bayesian optimization improved these values to  $R^2 = 0.999597$ , RMSE = 0.036338 nm, and MAE = 0.020013 nm. Because total thickness is both an input and a component of the derived response, this high score partly reflects a strong algebraic state-variable relationship and should not be interpreted as purely mechanistic prediction.

Table 4 reports the target-separated experimental-data comparisons. The mobile random forest reproduced all three tables closely. Landau–Levich was also strong for hexane but substantially overpredicted the other two systems, reaching 69.86 nm for toluene when the measured mobile range was 0–0.47 nm and 10.75 nm for ethyl acetate when the measured range was 0.26–3.18 nm.

Table 4: Experimental mobile-layer comparison. Landau–Levich is not included in bonded-thickness evaluation.

| Solvent       | Model                      | $R^2$         | RMSE (nm) | MAE (nm)  |
|---------------|----------------------------|---------------|-----------|-----------|
| Hexane        | Bayesian-optimized RF      | 0.999986      | 0.010386  | 0.006432  |
| Hexane        | Landau–Levich mobile layer | 0.990529      | 0.267725  | 0.211779  |
| Toluene       | Bayesian-optimized RF      | 0.930008      | 0.026170  | 0.011328  |
| Toluene       | Landau–Levich mobile layer | −95254.722098 | 30.529349 | 15.309791 |
| Ethyl acetate | Bayesian-optimized RF      | 0.999790      | 0.009323  | 0.006729  |
| Ethyl acetate | Landau–Levich mobile layer | −18.186017    | 2.821235  | 1.654359  |

The solvent dependence is informative rather than a reason to recalibrate the equation against each response. Hexane appears compatible with the assumed entrainment-plus-solids-conversion picture over the measured concentration range. The toluene and ethyl acetate failures indicate that fixed withdrawal conditions, pure-solvent property approximations, and concentration-only drying conversion are insufficient for those systems. The extreme toluene  $R^2$  is amplified by its narrow observed target variance, but its 30.53 nm RMSE independently confirms a large scale error. Missing effective evaporation, solution-property variation with concentration, interfacial adsorption, drainage, and solidification morphology are plausible contributors consistent with the analytical literature.

## 7 Experimental-only robustness follow-up

This analysis supersedes deployment interpretations of the preceding historical synthetic benchmarks; it does not replace their archived numerical results. All new scores use existing experimental records only. No independent new laboratory campaign has been supplied.

### 7.1 Audited data and leakage controls

The source tables contain 142 records. One toluene record has  $h_T = 0.85$  nm and  $h_B = 0.86$  nm and is quarantined without clipping. The first 15 toluene rows reproduce workbook summary values and may overlap individual readings; excluding them gives 126 primary records across 36 solvent–concentration conditions. A sensitivity analysis retains all 141 physically valid records.

The original concentration numbers are in g/L, as documented by the workbook unit labels; the canonical schema names them accordingly and provides  $c_{\text{g/mL}} = c_{\text{g/L}}/1000$ . Raw files are preserved, and source rows, unit-evidence cells and file hashes are archived.

The broader physical audit finds 192 of 3000 historical synthetic rows violating finite, non-negative concentration/thickness or  $h_T \geq h_B$  requirements. This is a broader criterion than the previously reported 113 negative mobile differences. A replacement training-only bootstrap perturbs nonnegative bonded and mobile components and reconstructs total as their sum. Its 300-row audit contains no invalid records and retains parent IDs. These synthetic rows are excluded from every validation experiment.

## 7.2 Methods

Bonded and mobile thickness are separate responses. Features are concentration and the four historical solvent descriptors. No measured thickness, including uncoated thickness, is used. Fixed models are the training mean, standardized Ridge regression ( $\alpha = 10$ ), and a 200-tree random forest with minimum leaf size two. Raw features are compared with  $\log(1 + c/(1 \text{ g/L}))$  and its interactions with solvent descriptors. Models are untuned; nonnegative projection is applied to predictions. Scaling is fitted only within training partitions.

Five-fold grouped cross-validation keeps each solvent-concentration condition intact. Leave-one-solvent-out tests train on two solvents and test on the third. Explicit low/high concentration tests reserve the lowest/highest  $\lceil 0.2K_s \rceil$  distinct concentrations in each solvent, where  $K_s$  is its number of represented conditions. These thresholds use features only. Batch and wafer IDs are unavailable, so condition groups are proxies and cannot rule out dependence across conditions prepared in the same batch. Grouped CV can include edge conditions and is not exclusively an interpolation experiment.

Point estimates use all outer-training rows. Separate interval estimators reserve 35% of outer-training groups for calibration and fit on the remainder. Each calibration score is the group’s maximum absolute residual. The nominal 80% split-conformal radius is order statistic  $\lceil (G+1)0.8 \rceil$  of the  $G$  group scores; an unavailable finite statistic produces an infinite radius. Interval centers therefore differ from full-training point predictions. Coverage relies on exchangeability and is not guaranteed under the deliberate distribution shifts evaluated here.

Group-weighted RMSE first averages squared residuals within each condition, then across conditions. Row-weighted RMSE, MAE and  $R^2$  are retained for comparison. Descriptive percentile ranges resample condition errors 2000 times and condition on the fitted folds; they do not measure retraining or solvent-population uncertainty. All fold assignments and individual predictions are committed.

### 7.3 Results

Table 5: Fixed raw-feature random forest on experimental records. RMSE is row-weighted; G-RMSE gives equal weight to each condition. Coverage is observed row coverage for separately calibrated nominal 80% intervals.

| Protocol           | Target | $R^2$  | RMSE  | G-RMSE | Coverage |
|--------------------|--------|--------|-------|--------|----------|
| Grouped CV         | Bonded | 0.568  | 0.106 | 0.106  | 89.7%    |
| Grouped CV         | Mobile | 0.879  | 0.705 | 0.945  | 87.3%    |
| Unseen solvent     | Bonded | -0.605 | 0.204 | 0.210  | 84.9%    |
| Unseen solvent     | Mobile | -0.456 | 2.445 | 2.928  | 71.4%    |
| Low concentration  | Bonded | -0.474 | 0.145 | 0.126  | 76.9%    |
| Low concentration  | Mobile | 0.051  | 0.173 | 0.152  | 100.0%   |
| High concentration | Bonded | 0.295  | 0.154 | 0.155  | 77.8%    |
| High concentration | Mobile | 0.157  | 3.243 | 4.157  | 14.8%    |

The RF grouped-CV scores are materially below the historical synthetic benchmark. Both pooled leave-one-solvent-out  $R^2$  values are negative. At high concentrations, mobile-layer row coverage falls to 14.8% and simultaneous condition coverage to 11.1%, despite the nominal 80% level. These results do not support reliable extrapolation or an exchangeability-based coverage guarantee in deployment.

Table 6: Leave-one-solvent-out raw RF point scores. Small within-solvent target variance can make  $R^2$  extreme, so absolute errors must also be considered.

| Solvent       | Target | $R^2$     | RMSE (nm) |
|---------------|--------|-----------|-----------|
| ethyl acetate | bonded | 0.217     | 0.232     |
| ethyl acetate | mobile | 0.248     | 0.559     |
| hexane        | bonded | -4.237    | 0.207     |
| hexane        | mobile | 0.009     | 2.739     |
| toluene       | bonded | -1.000    | 0.168     |
| toluene       | mobile | -1592.794 | 2.965     |

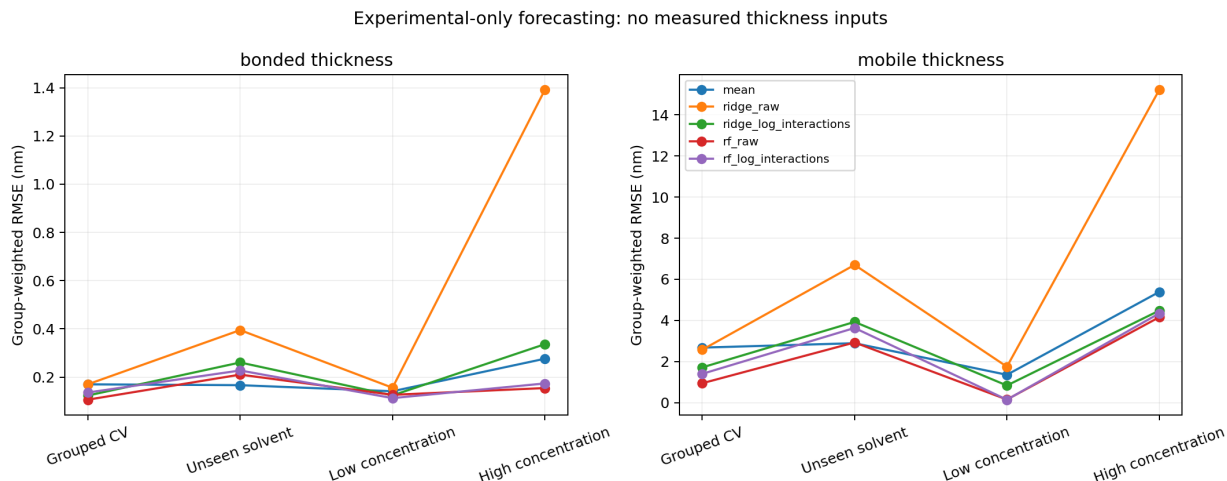


Figure 18: Experimental-only comparison across deployment scenarios. All fixed candidates, including the training-mean baseline, are shown.

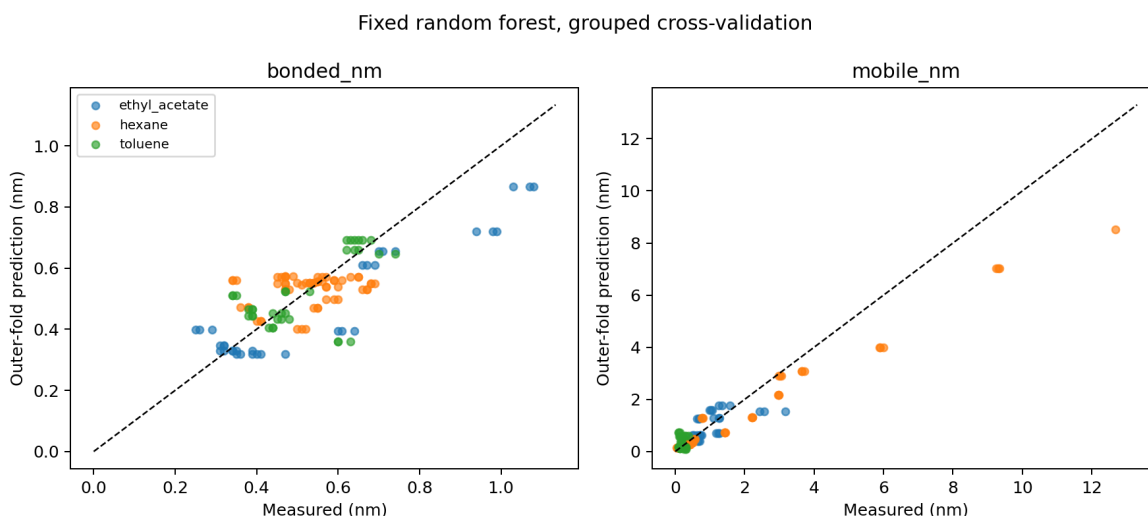


Figure 19: Grouped out-of-fold raw RF predictions. No measured thickness is a feature.

The log/interactions RF has grouped-CV group-weighted RMSE of 0.136 nm for bonded and 1.401 nm for mobile thickness, versus 0.106 and 0.945 nm for raw RF. Transformations modestly reduce low-concentration errors but do not yield a consistent improvement across scenarios. They remain exploratory choices, not validated universal gains.

Summary-inclusion sensitivity results and all model-level scores are available in the committed CSVs. This sensitivity changes both condition coverage and fold composition, so its differences cannot be attributed solely to duplicated summary weighting. It is not independent validation.

## 7.4 Remaining limitations and external test requirement

The three solvents, incomplete batch lineage, summary/reading ambiguity, and fixed approximate solvent descriptors limit inference. Solution properties, withdrawal speed, dwell time and protocol variation are not resolved by concentration-only process information. The workbook inventory is historical material and cannot establish an untouched external test set. A prospective protocol and empty intake schema are provided; they require newly collected independent batches, matched measurements and uncertainty, source-cell provenance, and confirmation that neither outcomes nor samples informed prior model or generator development. The final model and scoring plan must be frozen before opening that test set. Consequently the manuscript remains a research draft, and external-validation completion is not claimed.

## 8 Detailed Analysis

### 8.1 The central methodological lesson

Validation design controls interpretation. Changing a sequential split to target-stratified cross-validation changes the question being answered; it does not resolve a deployment distribution shift. The experimental-only grouped and OOD tests show that high synthetic interpolation scores coexist with poor transfer to unseen solvents and concentration extremes. These failures are scientific results that constrain the model’s scope.

### 8.2 Why the original hexane-only workflow was insufficient

The original hexane-only experimental table was sparse. CTGAN and bootstrap-plus-Cholesky augmentation explored ways to reproduce that table’s structure, but correlation preservation does not create independent experimental information. Augmentation fitted before splitting can transmit source information across nominal folds. The new validation therefore uses experimental records only, and the constrained generator is limited to a training-only feasibility audit.

### 8.3 Why tree models dominated

Tree ensembles dominated because they handle heterogeneous descriptors, nonlinear interactions, and monotonic nonlinearities without requiring carefully chosen transformations. They are also comparatively robust to mixed scales and irregular distributions. These advantages became key once the descriptor space expanded from three process/thickness variables to seven mixed process-and-chemistry features.

### 8.4 Residual structure and future feature engineering

Residual diagnostics identified various systematic opportunities. High absolute residual correlation with polarity suggests a regime effect not fully linearized by current features. Residual dependence on surface tension indicates that capillarity-linked behavior still contains unresolved structure. For boiling point, a spline or piecewise representation may better capture evaporation-related effects.

Finally, positive skew and residual trends in total thickness suggest the utility of log or quantile transforms. These signals support future feature engineering between polarity, viscosity, and surface tension, as well as transformed versions of thickness variables.

## 8.5 Interpretation of final feature importance

The final optimized random forest placed the greatest weight on concentration, total thickness, and uncoated layer, with polarity serving as the strongest secondary solvent descriptor. This ranking is physically plausible. Concentration governs solids loading, total thickness and uncoated layer summarize film-state evolution, and the solvent descriptors modulate the chemistry and transport environment rather than dominating it directly.

## 8.6 What the analytical comparison adds

The analytical exercise changes the interpretation of model quality. It shows that a physically appropriate law can perform extremely well in one solvent and still fail under another set of transport and drying conditions. It also shows why target identity matters: a Landau–Levich curve can be a defensible mobile-layer baseline without being a bonded-layer model. The random forest offers empirical interpolation across the variables represented in the synthetic corpus, while the analytical equations expose which unmeasured mechanisms—especially withdrawal speed, local evaporation, concentration-dependent solution properties, and adsorption kinetics—should be added to future experiments.

## 8.7 Limitations

The historical benchmark is predominantly synthetic and its scores depend on the generator and split design. The experimental-only follow-up avoids synthetic augmentation in evaluation but remains small and incompletely documented at the batch level. The solvent-specific tables contributed to the historical generator, so their earlier comparison scores measure alignment with source experiments rather than independent external generalization. Neither augmentation nor a different split substitutes for new measurements.

The historical mobile-target audit found 113 of 3000 synthetic rows with  $h_T - h_B < 0$ . The follow-up’s broader physical audit flags 192 rows, including invalid concentrations, and provides a constrained training-only generator. Historical mobile models use total thickness as both a predictor and a term in the response definition; they concern measured-state estimation rather than pre-process forecasting. The new models exclude all measured-thickness features. The canonical data release corrects the legacy g/mL label to g/L and supplies an explicit conversion, while keeping original files unchanged for provenance.

The Landau–Levich comparison is intentionally uncalibrated but correspondingly approximate. Withdrawal speed is fixed at 1.0 mm/s, solvent properties are treated as constants rather than concentration-dependent solution properties, PDMS volume conversion assumes additive volumes and complete solvent removal, and no local effective evaporation rate is available. These assumptions are particularly consequential for the poor toluene and ethyl acetate results. The capillary-



only comparator remains symbolic until effective evaporation and coating-zone measurements are collected; the mixed capillary/entrainment equation is retained as literature background rather than as a bonded comparator.

The chemical scope is narrow: three solvents and one solute, PDMS. Solvent descriptors are historical approximations rather than experiment-specific solution measurements. The leave-one-solvent-out failures preclude a claim of reliable chemistry transfer. Batch lineage is incomplete, and a true external dataset remains unavailable. Newly measured independent batches and broader solvent coverage are required before deployment claims are made.

## 9 Conclusion

This investigation developed a complete machine-learning framework for bonded-thickness prediction in PDMS dip coating across hexane, toluene, and ethyl acetate solvent systems. The work began from a sparse hexane experimental dataset, progressed through two synthetic-data-generation stages, expanded the descriptor space using solvent properties, and culminated in full-dataset validation, Bayesian optimization, and a target-separated mobile-layer analytical comparison.

The experimental-only follow-up changes the central performance conclusion. A fixed RF without measured-thickness inputs achieves grouped-CV  $R^2 = 0.568$  for bonded and 0.879 for mobile thickness, but pooled solvent-transfer scores are negative and high-concentration uncertainty coverage fails. Feature transforms do not yield consistent gains. The historical synthetic scores (0.9734 bonded and 0.9996 mobile) remain development diagnostics with different features and validation dependencies; they are not directly comparable evidence of external forecasting. Landau–Levich remains a mobile-layer comparator and is not fitted or scored against bonded thickness.

The immediate priority is independent experimental evidence and complete sample/batch provenance. The repository now provides audited units, physical exclusions, leakage-controlled experimental splits, prediction intervals, full reproducible results and a prospective test protocol. New independent experiments and scientific review remain outstanding, so neither deployment readiness nor a finalized externally validated manuscript is claimed.

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